



PHYS121 Integrated Science-Physics

W4T3 Angular Momentum (part 2)

References:

- [1] David Halliday, Jearl Walker, Resnick Jearl, 'Fundamentals of Physics', (Wiley, 2018)
- [2] Doug Giancoli, 'Physics for Scientists and Engineers with modern physics', (Pearson, 2009)
- [3] Hugh D. Young, Roger A. Freedman, 'University Physics with Modern Physics', (Pearson, 2012)

And others specified when needed.



Review & Summary

Rolling Bodies For a wheel of radius R rolling smoothly,

$$v_{\text{com}} = \omega R, \quad (11-2)$$

where v_{com} is the linear speed of the wheel's center of mass and ω is the angular speed of the wheel about its center. The wheel may also be viewed as rotating instantaneously about the point P of the “road” that is in contact with the wheel. The angular speed of the wheel about this point is the same as the angular speed of the wheel about its center. The rolling wheel has kinetic energy

$$K = \frac{1}{2} I_{\text{com}} \omega^2 + \frac{1}{2} M v_{\text{com}}^2, \quad (11-5)$$

where I_{com} is the rotational inertia of the wheel about its center of mass and M is the mass of the wheel. If the wheel is being accelerated but is still rolling smoothly, the acceleration of the center of mass $\vec{a}_{\text{com}}$ is related to the angular acceleration α about the center with

$$a_{\text{com}} = \alpha R. \quad (11-6)$$

If the wheel rolls smoothly down a ramp of angle θ , its acceleration along an x axis extending up the ramp is

$$a_{\text{com},x} = - \frac{g \sin \theta}{1 + I_{\text{com}}/MR^2}. \quad (11-10)$$

Torque as a Vector In three dimensions, *torque* $\vec{\tau}$ is a vector quantity defined relative to a fixed point (usually an origin); it is

$$\vec{\tau} = \vec{r} \times \vec{F}, \quad (11-14)$$

where $\vec{F}$ is a force applied to a particle and $\vec{r}$ is a position vector locating the particle relative to the fixed point. The magnitude of $\vec{\tau}$ is

$$\tau = rF \sin \phi = rF_{\perp} = r_{\perp}F, \quad (11-15, 11-16, 11-17)$$

where ϕ is the angle between $\vec{F}$ and $\vec{r}$, $F_{\perp}$ is the component of $\vec{F}$ perpendicular to $\vec{r}$, and $r_{\perp}$ is the moment arm of $\vec{F}$. The direction of $\vec{\tau}$ is given by the right-hand rule.

11.3.2. A hollow cylinder is rotating about an axis that passes through the center of both ends. The radius of the cylinder is r . At what angular speed ω must the this cylinder rotate to have the same total kinetic energy that it would have if it were moving horizontally with a speed v without rotation?

a) $\omega = \frac{v^2}{2r}$

b) $\omega = \frac{v}{r}\sqrt{2}$

c) $\omega = \frac{v}{r}$

d) $\omega = \frac{v}{2r}$

e) $\omega = \frac{v^2}{r^2}$

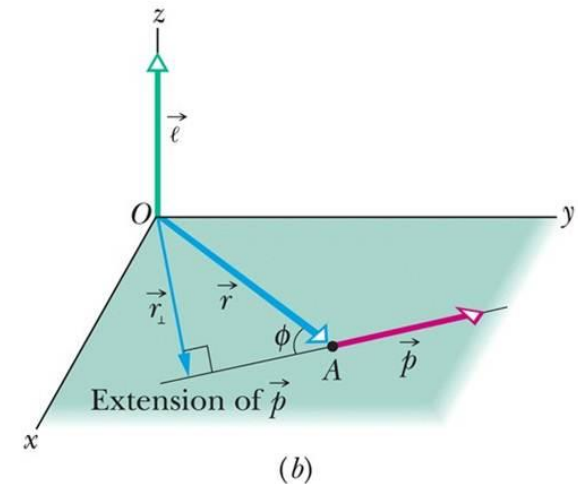
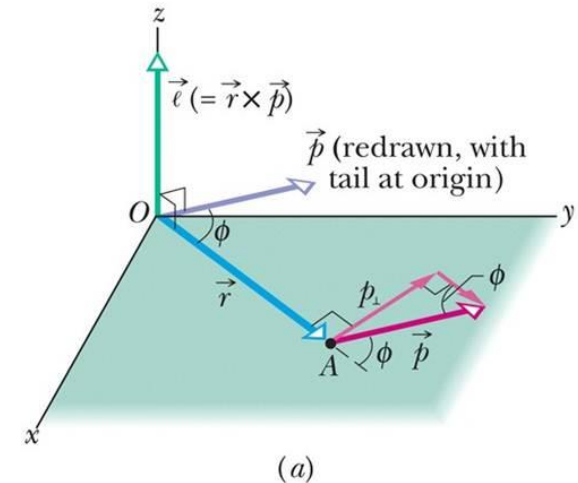


Learning Outcomes

- Find the angular momentum (a particle/an object) and apply Newton's second law for rotation.
- Apply conservation of angular momentum to solve problems (for one or more object).
- Describe and analyze precession motion.

Angular Momentum

- Here we investigate the angular counterpart to linear momentum
- We write: $\vec{\ell} = \vec{r} \times \vec{p} = m(\vec{r} \times \vec{v})$
- Note that the particle need not rotate around O to have angular momentum around it (O vs. the rotation axis)
- The unit of angular momentum is $\text{kg m}^2/\text{s}$, or J s



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Figure 11-12

- To find the direction of angular momentum, use the right-hand rule to relate r and v to the result
- To find the magnitude, use the equation for the magnitude of a cross product:

$$\ell = rmv \sin \phi, \quad \text{Equation (11-19)}$$

- Which can also be written as:

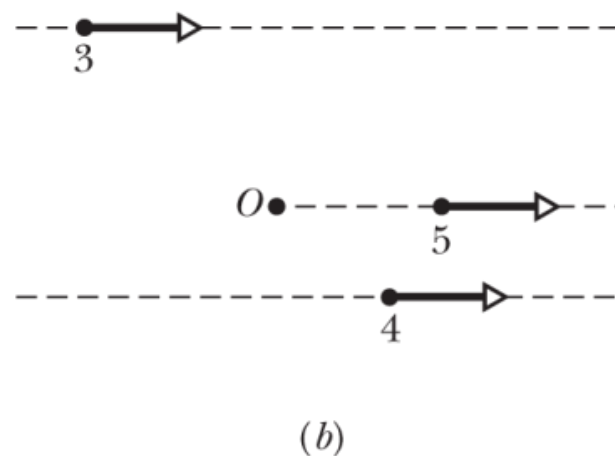
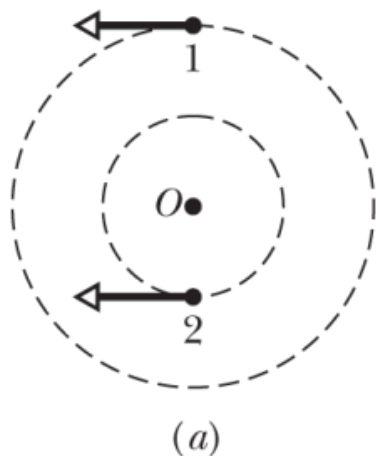
$$\ell = rp_{\perp} = rmv_{\perp}, \quad \text{Equation (11-20)}$$

$$\ell = r_{\perp} p = r_{\perp} mv, \quad \text{Equation (11-21)}$$

- *Angular momentum has meaning only with respect to a specified origin*
- It is always perpendicular to the plane formed by the position and linear momentum vectors

Checkpoint 4

In part *a* of the figure, particles 1 and 2 move around point *O* in circles with radii 2 m and 4 m. In part *b*, particles 3 and 4 travel along straight lines at perpendicular distances of 4 m and 2 m from point *O*. Particle 5 moves directly away from *O*. All five particles have the same mass and the same constant speed, (a) Rank the particles according to the magnitudes of their angular momentum about point *O*, greatest first, (b) Which particles have negative angular momentum about point *O*?



Answer:

(a) 1 & 3, 2 & 4, 5

(b) 2 and 3 (assuming counterclockwise is positive)

Newton's Second Law in Angular Form

- We rewrite Newton's second law as:

$$\vec{\tau}_{\text{net}} = \frac{d\vec{\ell}}{dt} \quad (\text{single particle}). \quad \text{Equation (11-23)}$$

- *The torque and the angular momentum must be defined with respect to the same point* (usually the origin)

The (vector) sum of all the torques acting on a particle is equal to the time rate of change of the angular momentum of that particle.

- Note the similarity to the linear form:

$$\vec{F}_{\text{net}} = \frac{d\vec{p}}{dt} \quad (\text{single particle}) \quad \text{Equation (11-22)}$$

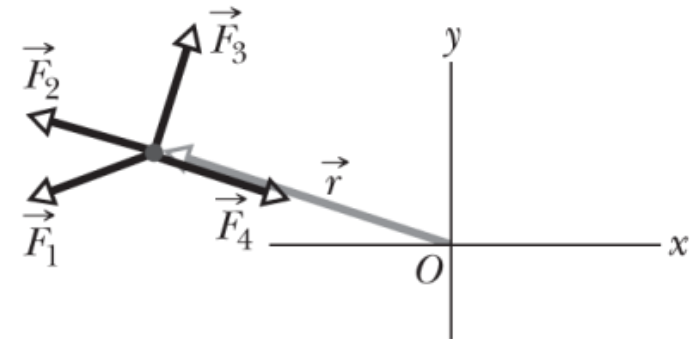
Checkpoint 5

The figure shows the position vector $\vec{r}$ of a particle at a certain instant, and four choices for the direction of a force that is to accelerate the particle. All four choices lie in the xy plane. (a) Rank the choices according to the magnitude of the time rate of change $\left(\frac{d\vec{\ell}}{dt}\right)$ they produce in the angular momentum of the particle about point O , greatest first, (b) Which choice results in a negative rate of change about O ?

Answer:

(a) F_3, F_1, F_2 & F_4

(b) F_3 (assuming counter clockwise is positive)



Angular Momentum of a Rigid Body

- We sum the angular momenta of the particles to find the angular momentum of a system of particles:

$$\vec{L} = \vec{\ell}_1 + \vec{\ell}_2 + \vec{\ell}_3 + \cdots + \vec{\ell}_n = \sum_{i=1}^n \vec{\ell}_i. \quad \text{Equation (11-26)}$$

- The rate of change of the net angular momentum is:

$$\frac{d\vec{L}}{dt} = \sum_{i=1}^n \vec{\tau}_{\text{net},i}. \quad \text{Equation (11-28)}$$

- In other words, the net torque is defined by this change:

$$\vec{\tau}_{\text{net}} = \frac{d\vec{L}}{dt} \quad (\text{system of particles}), \quad \text{Equation (11-29)}$$

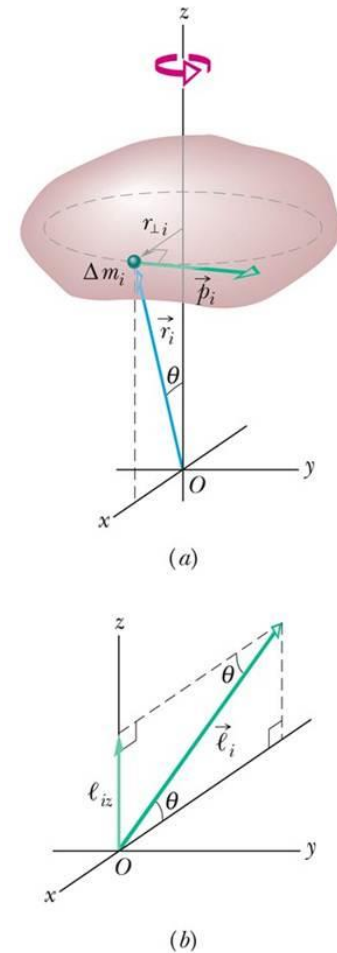
The net external torque $\vec{\tau}_{\text{net}}$ acting on a system of particles is equal to the time rate of change of the system's total angular momentum $\vec{L}$.

- Note that the torque and angular momentum must be measured relative to the same origin
- If the center of mass is accelerating, then that origin must be the center of mass
- We can find the angular momentum of a rigid body through summation:

$$\begin{aligned}
 L_Z &= \sum_{i=1}^n \ell_{iz} = \sum_{i=1}^n \Delta m_i v_i r_{\perp i} = \sum_{i=1}^n \Delta m_i (\omega r_{\perp i}) r_{\perp i} \\
 &= \omega \left(\sum_{i=1}^n \Delta m_i r_{\perp i}^2 \right).
 \end{aligned}$$

- The sum is the rotational inertia I of the body
- Therefore this simplifies to:

$$L = I\omega \quad (\text{rigid body, fixed axis}).$$



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Figure 11-15

Table 11-1 More Corresponding Variables and Relations for Translational and Rotational Motion^a

Translational	Translational	Rotational	Rotational
Force	$\vec{F}$	Torque	$\vec{\tau} (= \vec{r} \times \vec{F})$
Linear momentum	$\vec{p}$	Angular momentum	$\vec{\ell} (= \vec{r} \times \vec{p})$
Linear momentum ^b	$\vec{p} (= \sum \vec{p}_i)$	Angular momentum ^b	$\vec{L} (= \sum \vec{\ell}_i)$
Linear momentum ^b	$\vec{p} (= M \vec{v}_{\text{com}})$	Angular momentum ^c	$L = I\omega$
Newton's second law ^b	$\vec{F}_{\text{net}} = \frac{d\vec{p}}{dt}$	Newton's second law ^b	$\vec{\tau}_{\text{net}} = \frac{d\vec{L}}{dt}$
Conservation law ^d	$\vec{p} = \text{a constant}$	Conservation law ^d	$\vec{L} = \text{a constant}$

^aSee also Table 10-3.

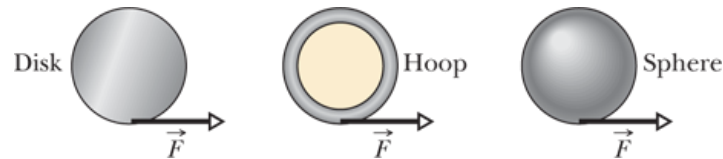
^bFor systems of particles, including rigid bodies.

^cFor a rigid body about a fixed axis, with L being the component along that axis.

^dFor a closed, isolated system.

Checkpoint 6

In the figure, a disk, a hoop, and a solid sphere are made to spin about fixed central axes (like a top) by means of strings wrapped around them, with the strings producing the same constant tangential force $\vec{F}$ on all three objects. The three objects have the same mass and radius, and they are initially stationary. Rank the objects according to (a) their angular momentum about their central axes and (b) their angular speed, greatest first, when the strings have been pulled for a certain time t .

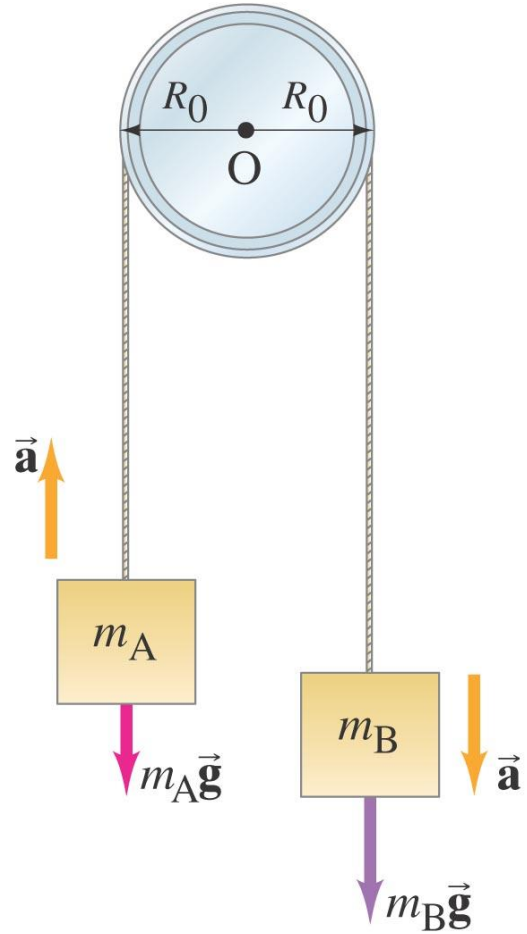


Answer:

- (a) All angular momenta will be the same, because the torque is the same in each case
- (b) sphere, disk, hoop

Example: Atwood's machine.

An Atwood machine consists of two masses, m_A and m_B , which are connected by an inelastic cord of negligible mass that passes over a pulley. If the pulley has radius R_0 and moment of inertia I about its axle, determine the acceleration of the masses m_A and m_B , and compare to the situation where the moment of inertia of the pulley is ignored.



Conservation of Angular Momentum

$$\vec{\tau}_{\text{net}} = \frac{d\vec{L}}{dt} \quad (\text{system of particles})$$

- Since we have a new version of Newton's second law, we also have a new conservation law:

$$\vec{L} = \text{a constant} \quad (\text{isolated system}). \quad \text{Equation (11-32)}$$

- The **law of conservation of angular momentum** states that, for *an isolated system*,

(net initial angular momentum) = (net final angular momentum)

$$\vec{L}_i = \vec{L}_f \quad (\text{isolated system}). \quad \text{Equation (11-33)}$$

If the net external torque acting on a system is zero, the angular $\vec{L}$ of the system remains constant, no matter what changes take place within the system.

- Since these are *vector equations*, they are equivalent to the three corresponding scalar equations
- This means we can separate axes and write:

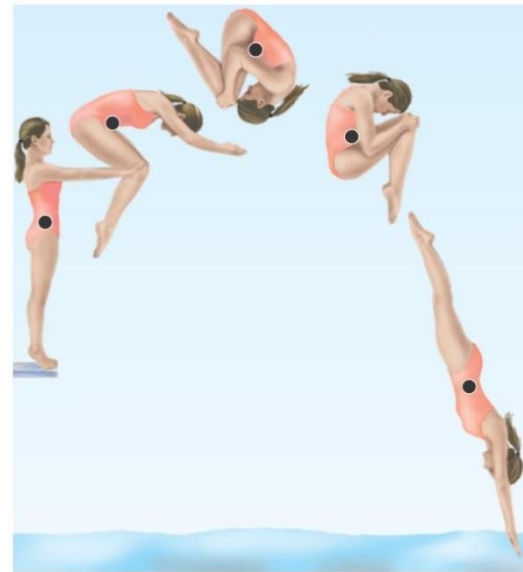
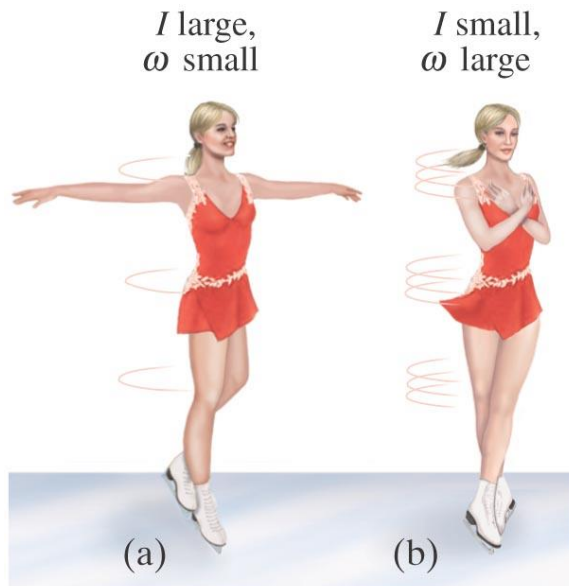
If the component of the net external torque on a system along a certain axis is zero, then the component of the angular momentum of the system along that axis cannot change, no matter what changes take place within the system.

- If the distribution of mass changes with no external torque, we have:

$$I_i \omega_f = I_f \omega_f . \qquad \textbf{Equation (11-34)}$$

Example Angular momentum conservation

- A student spinning on a stool: rotation speeds up when arms are brought in, slows down when arms are extended
- A springboard diver: rotational speed is controlled by tucking her arms and legs in, which reduces rotational inertia and increases rotational speed



Checkpoint 7

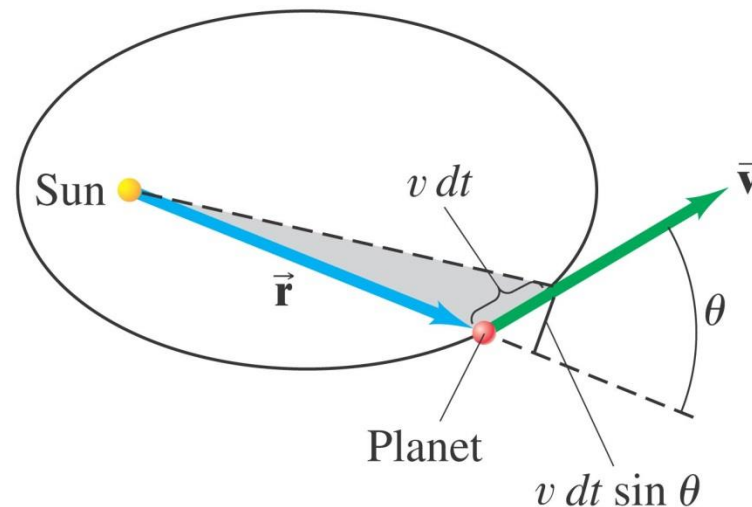
A rhinoceros beetle rides the rim of a small disk that rotates like a merry-go-round. If the beetle crawls toward the center of the disk, do the following (each relative to the central axis) increase, decrease, or remain the same for the beetle-disk system: (a) rotational inertia, (b) angular momentum, and (c) angular speed?

Answer:

- (a) decreases
- (b) remains the same
- (c) increases

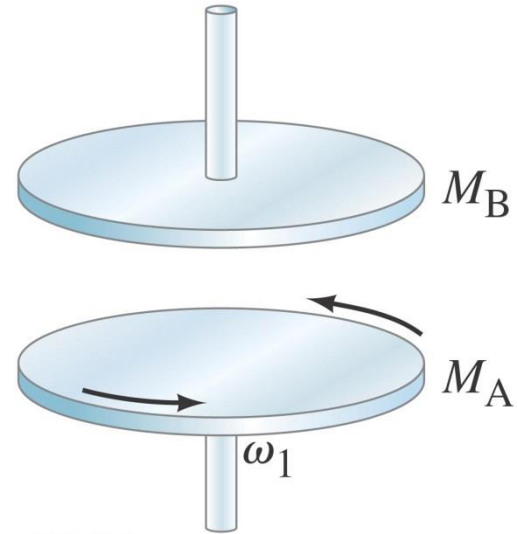
Example: Kepler's second law derived.

Kepler's second law states that each planet moves so that a line from the Sun to the planet sweeps out equal areas in equal times. Use conservation of angular momentum to show this.



Example: Clutch 离合器.

A simple clutch consists of two cylindrical plates that can be pressed together to connect two sections of an axle, as needed, in a piece of machinery. The two plates have masses $M_A = 6.0$ kg and $M_B = 9.0$ kg, with equal radii $R_0 = 0.60$ m. They are initially separated. Plate M_A is accelerated from rest to an angular velocity $\omega_1 = 7.2$ rad/s in time $\Delta t = 2.0$ s. Calculate (a) the angular momentum of M_A , and (b) the torque required to have accelerated M_A from rest to ω_1 . (c) Next, plate M_B , initially at rest but **free to rotate without friction**, is placed in **firm contact with freely rotating plate M_A** , and the two plates both rotate at a constant angular velocity ω_2 , which is considerably less than ω_1 . Why does this happen, and what is ω_2 ?

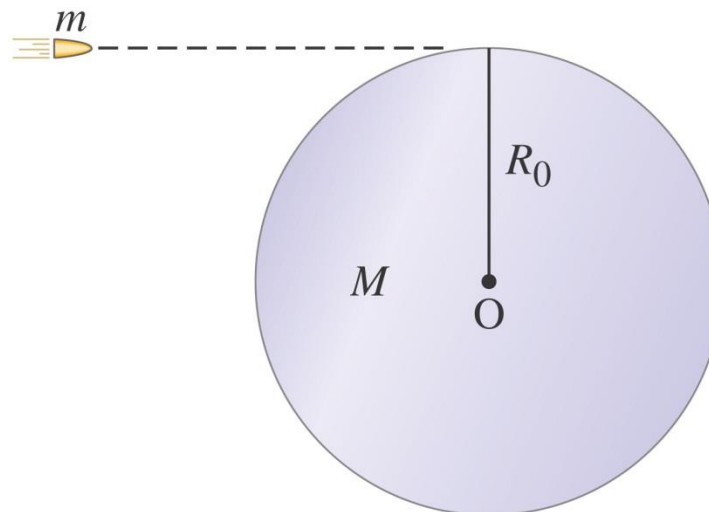


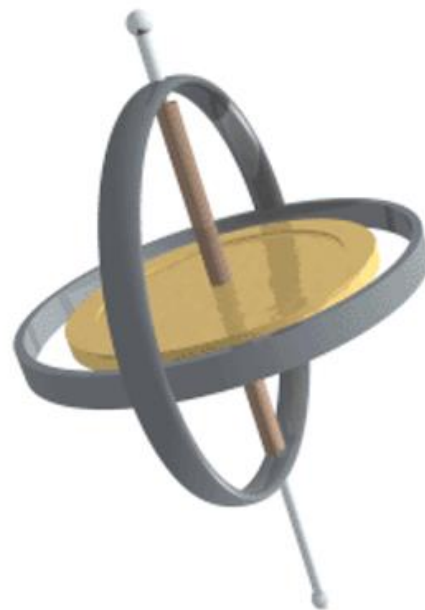
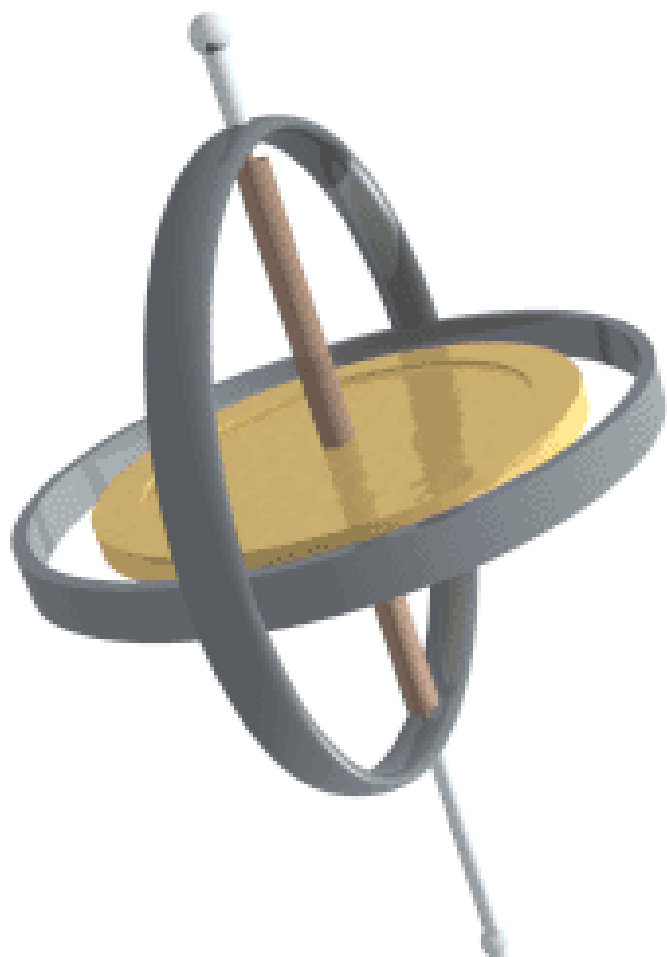


<https://cityclutchandbrake.co.za/clutch-and-pressure-plate-reconditioning/>

Example: Bullet strikes cylinder edge.

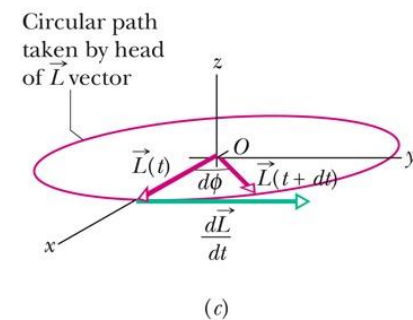
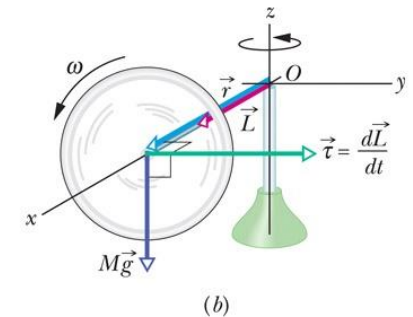
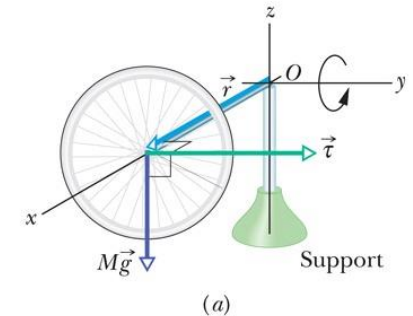
A bullet of mass m moving with velocity v strikes and becomes **embedded** at the edge of a cylinder of mass M and radius R_0 . The cylinder, **initially at rest**, begins to rotate about its symmetry axis, which **remains fixed** in position. Assuming **no frictional torque**, what is the angular velocity of the cylinder after this collision? Is kinetic energy conserved?





Precession of a Gyroscope

- A nonspinning gyroscope, as attached in 11-22 (a), falls
- A spinning gyroscope (b) instead rotates around a vertical axis
- This rotation is called **precession**



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Figure 11-22

- The angular momentum of a (rapidly spinning) gyroscope is:

$$L = I\omega, \quad \text{Equation (11-43)}$$

- The torque can only change the direction of L , not its magnitude, because of (11-43)

$$d\vec{L} = \vec{\tau}dt. \quad \text{Equation (11-44)}$$

- The only way its direction can change along the direction of the torque without its magnitude changing is if it rotates around the central axis
- Therefore it precesses instead of toppling over

$$dL = \tau dt = Mgr dt.$$

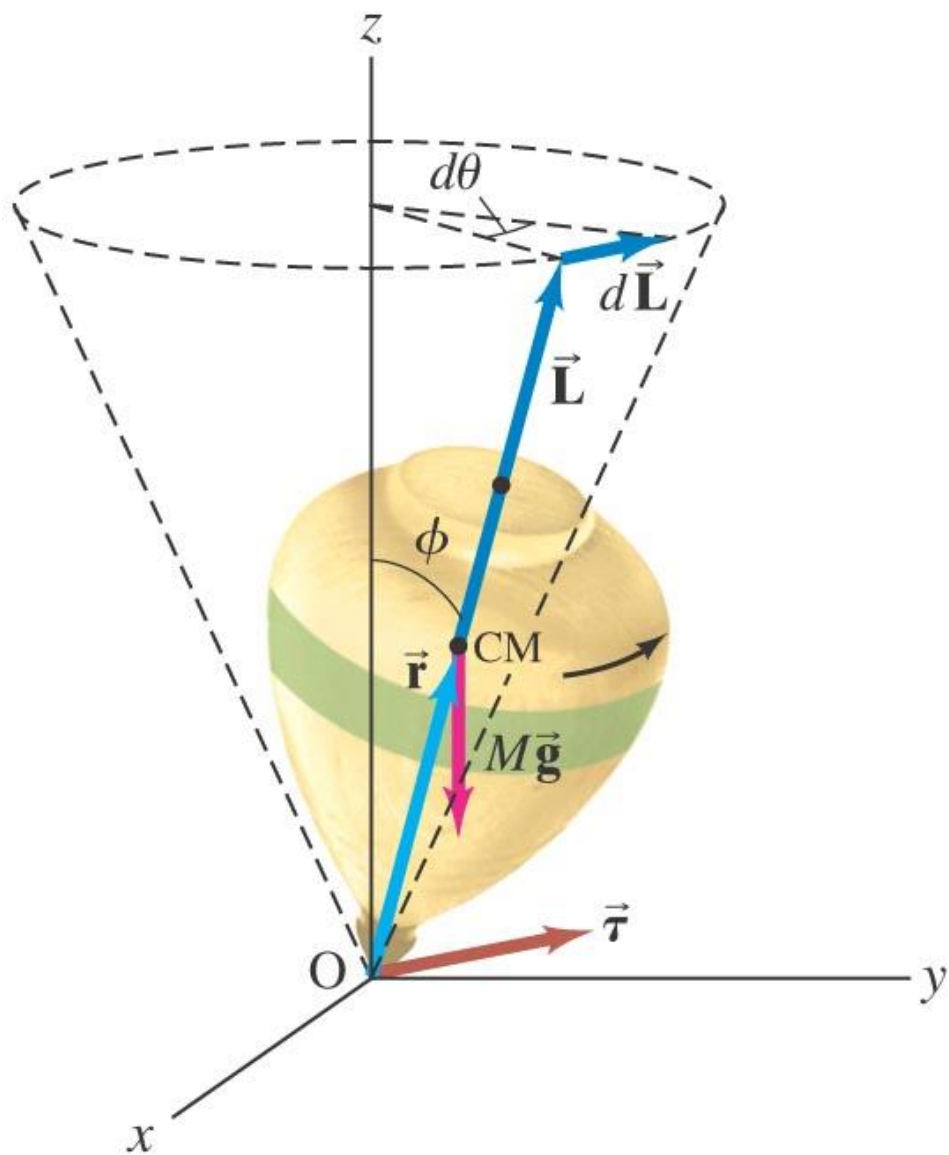
$$d\phi = \frac{dL}{L} = \frac{Mgr dt}{I\omega}.$$

- The **precession rate** is given by:

$$\Omega = \frac{Mgr}{I\omega}$$

Equation (11-46)

- True for a sufficiently rapid spin rate
- Independent of mass, (I is proportional to M) but does depend on g
- Valid for a gyroscope at an angle to the horizontal as well (a top for instance)





Learning Outcomes

- ✓ Find the angular momentum (a particle/an object) and apply Newton's second law for rotation.
- ✓ Apply conservation of angular momentum to solve problems (for one or more object).
- ✓ Describe and analyze precession motion.

$$\vec{\ell} = \vec{r} \times \vec{p} = m(\vec{r} \times \vec{v}) \quad \vec{\tau}_{\text{net}} = \frac{d\vec{\ell}}{dt} \quad \vec{\tau}_{\text{net}} = \frac{d\vec{L}}{dt}$$

$$L = I\omega$$

$$\vec{L}_i = \vec{L}_f$$

$$\Omega = \frac{Mgr}{I\omega}$$



CHAPTER REVIEW AND EXAMPLES, DEMOS

<https://openstax.org/books/university-physics-volume-1/pages/11-summary>

Even if the particle is not rotating about the origin, we can still define an angular momentum in terms of the position vector and the linear momentum.

... In this respect, the magnitude of the angular momentum depends on the choice of origin.

Celestial objects such as planets have angular momentum due to their **spin** and **orbits** around stars.

FIGURE 11.2



(a)



(b)

- (a) The bicycle moves forward, and its tires do not slip. The bottom of the slightly deformed tire is at rest with respect to the road surface for a measurable amount of time.
- (b) This image shows that the top of a rolling wheel appears blurred by its motion, but the bottom of the wheel is instantaneously at rest. (credit a: modification of work by Nelson Lourenço; credit b: modification of work by Colin Rose)

Example 11.6

Angular Momentum of a Robot Arm

A robot arm on a Mars rover like *Curiosity* shown in **Figure 11.8** is 1.0 m long and has forceps at the free end to pick up rocks. The mass of the arm is 2.0 kg and the mass of the forceps is 1.0 kg. See **Figure 11.13**. The robot arm and forceps move from rest to $\omega = 0.1\pi$ rad/s in 0.1 s. It rotates down and picks up a Mars rock that has mass 1.5 kg. The axis of rotation is the point where the robot arm connects to the rover. (a) What is the angular momentum of the robot arm by itself about the axis of rotation after 0.1 s when the arm has stopped accelerating? (b) What is the angular momentum of the robot arm when it has the Mars rock in its forceps and is rotating upwards? (c) When the arm does not have a rock in the forceps, what is the torque about the point where the arm connects to the rover when it is accelerating from rest to its final angular velocity?

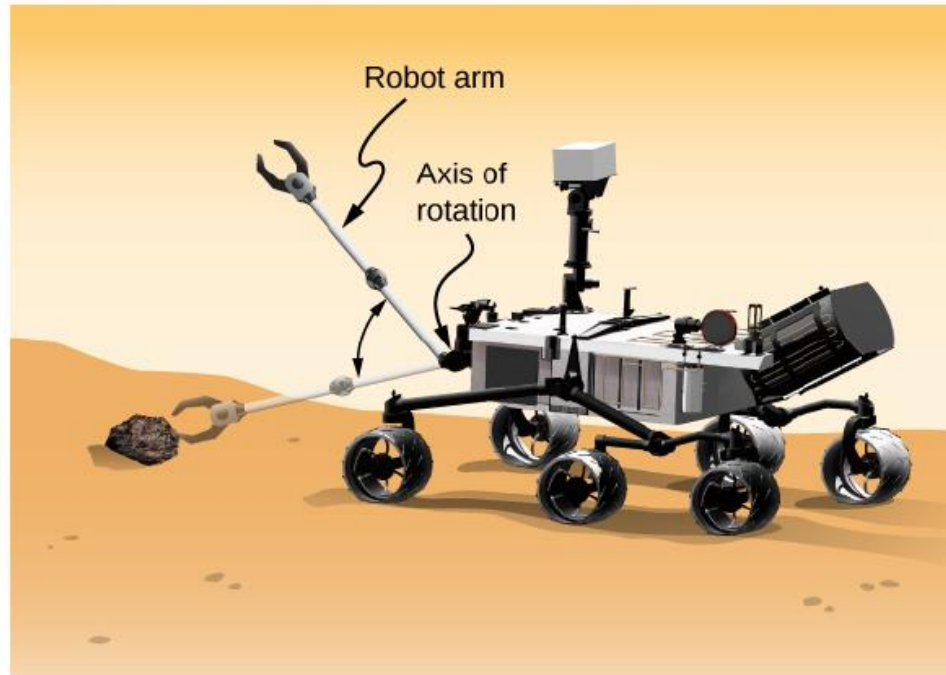


Figure 11.13 A robot arm on a Mars rover swings down and picks up a Mars rock. (credit: modification of work by NASA/JPL-Caltech)

Example 11.8

Dismount from a High Bar

An 80.0-kg gymnast dismounts from a high bar. He starts the dismount at full extension, then tucks to complete a number of revolutions before landing. His moment of inertia when fully extended can be approximated as a rod of length 1.8 m and when in the tuck a rod of half that length. If his rotation rate at full extension is 1.0 rev/s and he enters the tuck when his center of mass is at 3.0 m height moving horizontally to the floor, how many revolutions can he execute if he comes out of the tuck at 1.8 m height? See **Figure 11.17**.

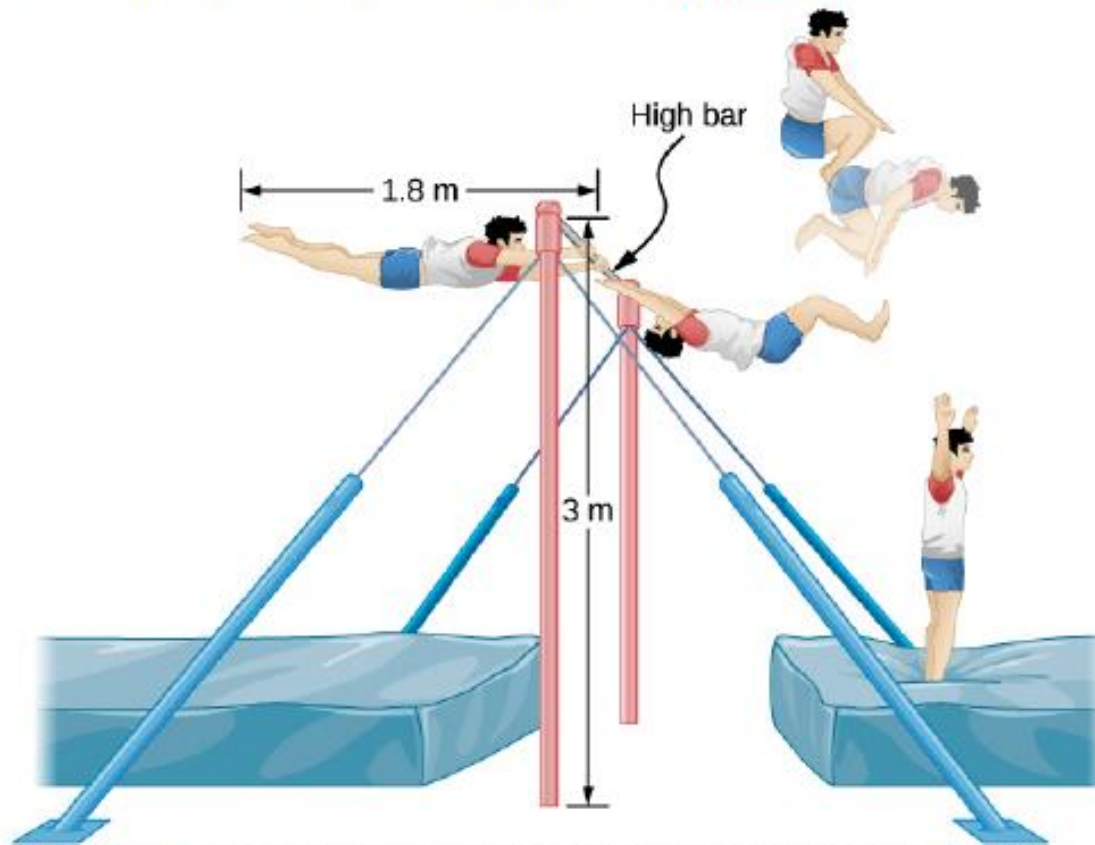


Figure 11.17 A gymnast dismounts from a high bar and executes a number of revolutions in the tucked position before landing upright.

Questions

